Assignment :- 5

Date: / /
Page No.

① Define Big Data and its characteristics.

→ Big Data refers to extremly large dataset that traditional systems can't handle. Its main characteristic are volume, velocity, variety, veracity, and value.

② What are the three main components of Hadoop?

→ HDFS for storage, YARN for resource management, and MapReduce for data processing.

③ Explain the mapReduce programming model.

→ It splits tasks into map (data processing) and Reduce (aggregation) phases.
Enables parallel processing across distributed nodes.

④ Differentiate between NameNode and DataNode in HDFS

→ NameNode manages metadata and file structure.
DataNode store actual data blocks and handle read/write operations.

⑤ What is the purpose of YARAN in Hadoop?

→ YARAN (Yet Another Resource Negotiator) manages cluster resources and job scheduling. It allows multiple processing frameworks to run on Hadoop.

⑥ Describe the Bulk Synchronous parallel (BSP) model

→ BSP divides computation into supersteps with synchronization barriers. used in parallel and graph processing frameworks like Giraph.

⑦ What are the advantages of HDFS?

→ Provides fault tolerance, scalability, and high throughput for large datasets.
Data replication ensures reliability even if nodes fail.

⑧ Explain the role of combiners in MapReduce.

→ combines perform local aggregation of map output before sending to reducers. They reduce network congestion and improve performance.

⑨ What is Pregel and how does it process graph?

→ Pregel is a vertex-centric graph processing model by google.
Each vertex performs computations and exchanges message in supersteps.

⑩ Describe the PageRank algorithm.

→ Rank web pages based on the no and quality of incoming links. higher-ranked pages contribute more weight to others they link to.

Date: / /
Page No.

(11) What are the limitations of MapReduce?

→ High disk I/O, limited iterative processing, and poor real-time performance.
Not ideal for low-latency or graph-based computations.

(12) Explain the concept of 'superstep' in graph processing.

→ Each superstep involves computation at vertices and message passing. After synchronization, the next superstep begins.

(13) What is the default replication factor in HDFS?

→ The default replication factor is 3, meaning each data block is stored three times.

(14) Differentiate between structured and unstructured Big Data.

→ Structured data follows a defined schema (tables, rows).
Unstructured data includes text, images, and videos without a fixed format.

(15) Describe the shuffle phase in MapReduce.

→ It transfers and group's intermediate data from mappers to reducers.
Ensures that all values for a key are sent to the same reducer.

Date: / /
Page No.

(17) Explain the role of Resource management in YARN.

→ It manages cluster resources, schedules jobs, and assigns containers to applications.

(18) What is speculative execution in Hadoop?

→ Runs duplicate tasks on slow nodes to speed up job completion. The first finished task's result is accepted, improving reliability.

(19) Describe the HBase database.

→ NoSQL column-oriented database built on top of HDFS.
Provides real-time read/write access to large ~~databa~~ datasets.

(20) What are the benefits of sequencefile format?

→ Stores key-value pairs in a binary format for efficiency.
Supports compression and fast data serialization.

(21) Explain the concept of input splits in MapReduce.

→ Divides input data into chunks for parallel processing by mapper tasks.
Each split is typically processed by one map task.

Date: / /
Page No.

(22) What is the role of Zookeeper in Hadoop?

→ Provides distributed coordination, synchronization, and configuration management.
Ensures reliable communication between Hadoop components.

(23) Describe the Graph framework.

→ An open source implementation of pregel for large-scale graph processing on Hadoop.

(24) What are the advantages of columnar storage for Big Data?

→ Faster analytical queries, better compression, and reduced I/O.
Ideal for aggregation-heavy workloads.

(25) Explain the concept of data locality in Hadoop.

→ Brings computation closer to where the data resides.
Reduces network traffic and boosts performance.

(26) What is the purpose of JobTracker in Hadoop?

→ In ~~older~~ older Hadoop versions, it scheduled MapReduce jobs and monitored progress.
Now replaced by YARN's Resource Manager.

Date: / /
Page No.

(27) Describe the triangle counting algorithm for graphs.

→ Count sets of three interconnected vertices to find dense clusters.
Used in social network and community analysis.

(28) What are the components of a MapReduce job?

→ Mapper, Reducer, Combiner, Partitioner, and Input / output formats.

(29) Explain the concepts of block size in HDFS.

→ HDFS divides files into fixed-size blocks (default 128 MB). Large block sizes improves throughput for big files.

(30) What is the role of TaskTracker in Hadoop?

→ Executes tasks on DataNodes and reports progress to the JobTracker. Replaced by NodeManager in YARN.

(31) Describe the connected components algorithms.

→ Identifies groups of vertices connected by paths. Used for network analysis and community detection.

(32) What are the benefits of in-memory processing frameworks?

→ Faster computation by avoiding disk I/O.
Frameworks like Apache Spark perform iterative tasks efficiently.

(33) Explain the concept of vertex-centric graph processing.

→ Each vertex independently performs computation and communicates with neighbours.
Used in models like Pregel and Giraph.

(34) What is the purpose of Avro file format?

→ A row-based data serialization format with rich schema support.
Used for efficient data exchange in Hadoop.

(35) Describe the betweenness centrality algorithm.

→ Measures how often a vertex lies on shortest path between others.
Identifies influential ~~tools~~ nodes in networks.

(36) What are the challenges in processing large graphs?

→ High memory use, communication overhead, and data partitioning issues.
Require distributed / parallel frameworks for scalability.

Date: / /
Page No.

(37) Explain the role of secondary NameNode.

→ Performs periodic checkpoints of the NameNode's metadata.
Helps with recovery but doesn't replace the main NameNode.

(38) What is the purpose of partitioning in MapReduce?

→ Divides intermediate data among reducers based on key values.
Ensures balanced workload distribution.

(39) Describe the GraphX framework.

→ Part of Apache Spark, used for in-memory graph computation.
Combines graph and data - parallel processing efficiently.

(40) What are the advantages of NoSQL database for Big Data?

→ Handle unstructured data, scale horizontally, and offer high performance. Provide flexible schema and fast read/write operations.

12/11/25